

Q<sub>1</sub>) จุด B(1,0,0) ไป A(0,2,0) สนามไฟฟ้า  $E = 5a_x \text{ V/m}$  มีประจุ 4C

$$\frac{\Delta y}{\Delta x} = \frac{y_A - y_B}{x_A - x_B} = \frac{y - y_B}{x - x_B}$$

$$\frac{y - y_B}{y_A - y_B} = \frac{x - x_B}{x_A - x_B}$$

$$\frac{y - 0}{2 - 0} = \frac{x - 1}{0 - 1}$$

$$x - 1 = -\frac{y}{2}$$

$$x = 1 - \frac{y}{2}$$

$$W = - \int_B^A \vec{E} \cdot d\vec{L}$$

$$\therefore W = q \int_B^A \vec{E} \cdot d\vec{L}$$

$$W = q \int_B^A E_x dx$$

$$W = 4 \int_1^0 5 dx$$

$$W = 4 [5x]_1^0$$

$$= 4(-5)$$

$$= -20 \text{ Joules}$$